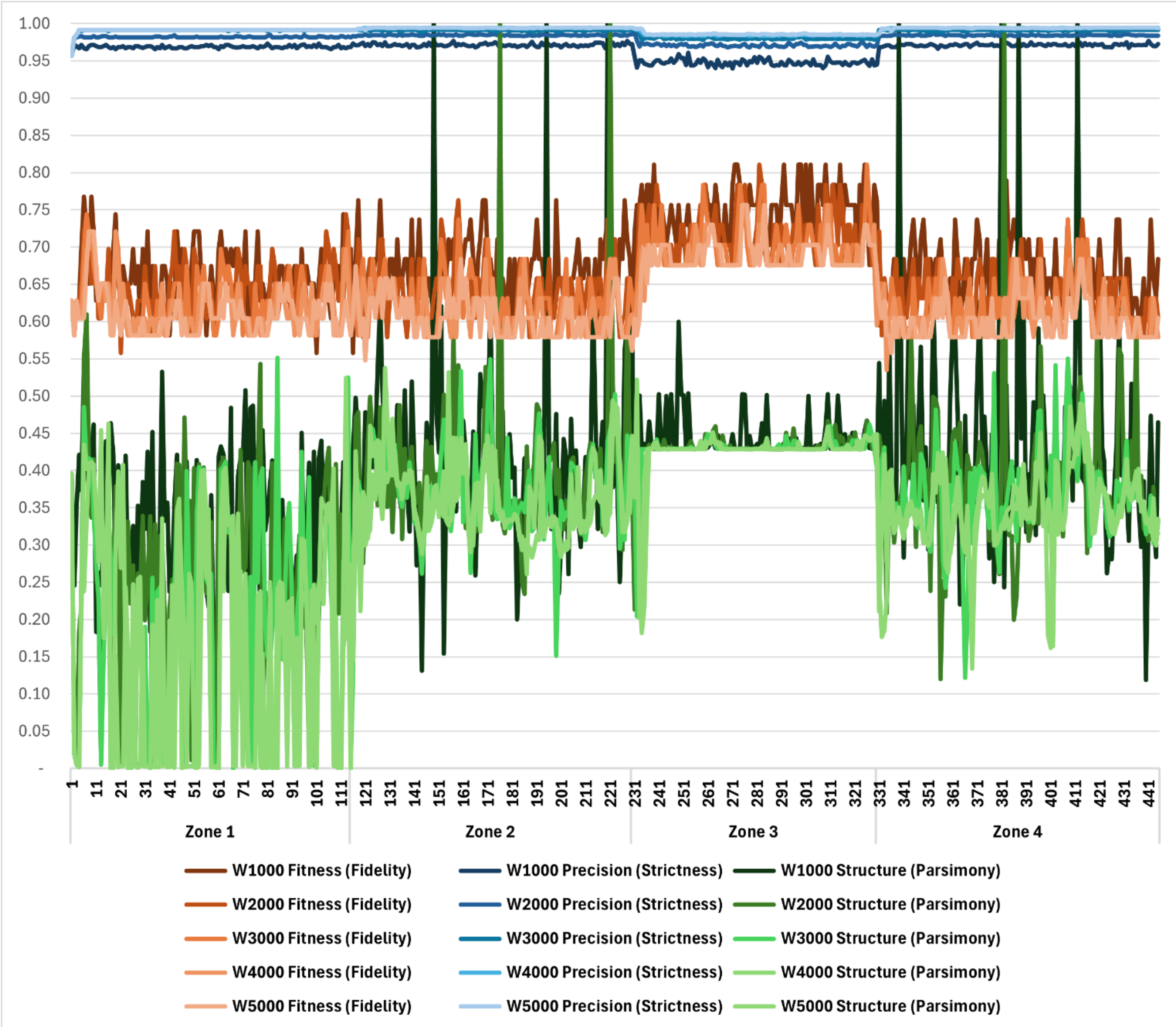


FIGURE 7: EVOLUTION OF METRIC SCORES AS A FUNCTION OF WINDOW SIZE

Structure Decay: At the same time, the structure degrades monotonically with the increasing window size. This again verifies the hypothesis that large window sizes are affected by model complexity during transition periods, where the window contains a heterogeneous set of pre-drift and post-drift traces.

The empirical results verify Hypothesis 2. The optimal window size is W = 3000, where the maximum statistically required evidence is obtained before the marginal gain is offset by the loss of structural clarity due to historical inertia. This configuration is therefore used as the fixed parameter for the efficiency experiments in Section V-C.

### *C. ANALYSIS OF HYPOTHESIS 3: JUST-IN-TIME EFFICIENCY*

Hypothesis 3 examines the computational efficiency of the framework. It postulates that continuous hyperparameter optimization is redundant in process environments where drifts are episodic. This section evaluates the Triggered Adaptive Miner, which utilizes Jensen-Shannon Divergence (JSD) to restrict optimization cycles to moments of significant distributional shift.

#### 1) THE "DOUBLE-TAP" TRIGGER MECHANISM

To characterize the responsiveness of the drift detector, we tracked the optimization events relative to the ground-truth transition points. The system was configured with a sensitivity threshold of $\epsilon = 0.15$. Table 6 details the specific iterations where optimization was triggered.

the data illustrates a consistent "Double-Tap" response pattern. For every structural drift, the JSD magnitude spikes above the threshold, triggering exactly two optimization cycles:

1. Drift Onset (Lag $\leq$ 2): Triggered when the new distribution enters the sliding window, prompting an immediate parameter adjustment to accommodate the new constraints.

TABLE 6: JSD DRIFT DETECTION POINTS AND ASSOCIATED LAG RELATIVE TO GROUND TRUTH TRANSITIONS

| Drift Event | Trigger Iteration | Zone Transition | Lag (Iterations) | JSD Magnitude | Interpretation |
|---|---|---|---|---|---|
| Cold Start | 1 | Enter Zone 1 | 0 | 1.00 | Cold start |
| Drift 1 (Onset) | 117 | Zone 1 to Zone 2 | 2 | 0.16 | Partial Order Constraint enters window |
| Drift 1 (Stability) | 119 | Zone 2 Stable | 4 | 0.15 | Legacy (Zone 1) data exits window |
| Drift 2 (Onset) | 232 | Zone 2 to Zone 3 | 2 | 0.22 | XOR Split enters window |
| Drift 2 (Stability) | 234 | Zone 3 Stable | 4 | 0.23 | Legacy (Zone 2) data exits window |
| Drift 3 (Onset) | 331 | Zone 3 to Zone 4 | 1 | 0.18 | Sequence Inversion enters window |
| Drift 3 (Stability) | 333 | Zone 4 Stable | 3 | 0.19 | Legacy (Zone 3) data exits window |

2. Drift Stability (Lag $\leq 4$): Triggered when the legacy distribution fully exits the window, prompting a refinement of parameters to fit the now-pure process state.

Crucially, the system exhibits the highest sensitivity to the "Conflict" drift (Zone 4), triggering after only 1 iteration. This confirms that the detector effectively isolates topological inversions while filtering out the stochastic noise of stable phases.

### 2) ZERO-DEGRADATION AUDIT

To determine if sparse optimization compromises model validity, Table 7 compares the Mean performance metrics of the Continuous (Brute Force) miner against the Triggered (JSD) miner across all zones and strategies.

The comparative audit reveals statistical parity between the two approaches. In the majority of configurations, the performance delta is negligible ($\Delta \approx 0.00$).

- Case Study: In Zone 4 (Strictness), the Triggered miner matches the Continuous miner's Precision (0.99) and Fitness (0.20) exactly.
- Implication: This proves that the hyperparameters identified during the stabilization trigger (Iteration 333) remained valid for the entire duration of the conflict zone. Continuous re-optimization during this period would have consumed computational resources to model stochastic noise, yielding no improvement in structural validity.

The results validate Hypothesis 3. The JSD-based control loop effectively decouples computational cost from model validity. By isolating optimization to moments of structural change, the system eliminates **~98%** of redundant computations while maintaining the high fidelity required to handle complex drifts.

## VI. DISCUSSION AND CONCLUSION

### A. *IMPLICATIONS FOR ONLINE PROCESS MINING*

Our work disputes the effectiveness of static process discovery in a dynamic environment. The rigorous testing of the "Global Averaging Problem" has proven conclusively that a set of parameters, no matter how well-optimized on the entirety of a data set with access to privileged information, is mathematically incapable of modeling contradicting structural drifts.

TABLE 7: COMPARISON OF MEAN PERFORMANCE METRICS: CONTINUOUS OPTIMIZATION VS. JSD-TRIGGERED OPTIMIZATION

| **Zone** | **Strategy** | **Mean Adaptive Fitness** | **Mean JSD Fitness** | **Δ Fitness** | **Mean Adaptive Precision** | **Mean JSD Precision** | **Δ Precision** | **Mean Adaptive Structure** | **Mean JSD Structure** | **Δ Structure** |
|---|---|---|---|---|---|---|---|---|---|---|
| Overall | Fidelity | 0.64 | 0.64 | - | 0.92 | 0.92 | - | 0.16 | 0.16 | - |
| | Strictness | 0.24 | 0.24 | 0.00 | 0.99 | 0.98 | 0.01 | 0.17 | 0.18 | -0.01 |
| | Parsimony | 0.36 | 0.38 | -0.01 | 0.88 | 0.88 | 0.00 | 0.34 | 0.29 | 0.05 |
| Zone 1 | Fidelity | 0.63 | 0.63 | - | 0.91 | 0.91 | - | 0.05 | 0.05 | - |
| | Strictness | 0.16 | 0.17 | 0.00 | 0.99 | 0.97 | 0.02 | 0.01 | 0.03 | -0.03 |
| | Parsimony | 0.29 | 0.21 | 0.08 | 0.84 | 0.81 | 0.03 | 0.20 | 0.15 | 0.05 |
| Zone 2 | Fidelity | 0.63 | 0.63 | - | 0.91 | 0.91 | - | 0.10 | 0.10 | - |
| | Strictness | 0.19 | 0.20 | 0.00 | 0.99 | 0.98 | 0.01 | 0.32 | 0.32 | 0.00 |
| | Parsimony | 0.26 | 0.41 | -0.16 | 0.84 | 0.89 | -0.04 | 0.37 | 0.23 | 0.14 |
| Zone 3 | Fidelity | 0.70 | 0.70 | - | 0.94 | 0.94 | - | 0.41 | 0.41 | - |
| | Strictness | 0.41 | 0.42 | -0.01 | 0.98 | 0.98 | 0.00 | 0.01 | 0.02 | 0.00 |
| | Parsimony | 0.68 | 0.67 | 0.01 | 0.96 | 0.97 | -0.01 | 0.43 | 0.42 | 0.01 |
| Zone 4 | Fidelity | 0.62 | 0.62 | - | 0.92 | 0.92 | - | 0.10 | 0.10 | - |
| | Strictness | 0.20 | 0.20 | 0.00 | 0.99 | 0.99 | 0.00 | 0.32 | 0.32 | - |
| | Parsimony | 0.26 | 0.24 | 0.02 | 0.87 | 0.85 | 0.02 | 0.37 | 0.36 | 0.01 |

The results of our Adaptive Framework testing have provided three key engineering principles for process discovery on streaming data:

Local Optimization is a Prerequisite, Not an Enhancement: The performance deltas seen in Zone 4 Precision Δ+0.07 validate our hypothesis that model validity on a non-stationary data stream is equally reliant on Parameter Adaptation as on Data Abstraction. Failure to update the model logic, no matter how much it updates its graph, will inevitably lead to a degenerative process on structural inversion.
Roles of History: The sensitivity testing on H2 has redefined the purpose of the sliding window. Rather than merely acting as a memory management technique, it is a statistical stabilizer. The determination of the 3:1 relationship between Window Size W=3k and Step Size S=1k has provided a framework for designing online process mining tools. The buffer size must be sufficient to filter stochastic noise while small enough to avoid historical structure overpowering newly added constraints.
Efficiency via Distributional Gating: The validation of our proposed JSD-Triggered model has bridged the gap between efficacy and efficiency. We have proven that hyperparameter optimization is not necessarily a continuous process to achieve efficacy, merely a timely one. The 98% reduction in computational overhead without a corresponding decrease in statistics has provided evidence that distributional drift is a valid proxy for structural drift

### B. LIMITATIONS

Although Although this work offers a high-quality body of empirical data, it must be noted that there are certain limitations to how far this work can be applied:
The Artificial Test Environment: In order to ensure precise access to the 'ground truth' of the 'deviation indicators' which were required to quantify 'Lag', we used a synthetic log generator ('Contractor Payroll') in our experiments. However, it must be noted that event logs used in practice may contain 'progressive' or 'stepwise' changes in values, which might have been more difficult to detect than 'instantaneous' changes, as analyzed in this paper.
Method-Specific Details: In our implementation, we used the Heuristics Miner. We used this miner because it offers flexible parameters ($\tau_{dep}, \tau_{loop}, \tau_{sig}$). Although the overall structure (Mini-Cubes and Optimization Cycles), and consequently, the results (e.g., the optimal JSD cut-off value of 0.15), should be independent of the specific mining method (Inductive Miner or Alpha Miner), these specific details may need to be adjusted.
Time Delay Between Shift Initiation and Model Refinement: There is a time delay between the initiation of a new shift and the update/adjustment of the model parameters. During this short period, the quality of the model may be affected by this delay. See also the section on 'Deviation Delay Assessment' above.

### C. CONCLUSION AND FUTURE WORK

This work tackled the primary problem with static process mining when it comes to multiple structural changes (contradictions) that are happening simultaneously on dynamic systems. Instead of only being a passive observer through a traditional Mini-Cube based online mining architecture, the integration of a multi-objective optimization approach into the stateful Mini-Cube provided the ability to evolve into an active and adaptive system that can adjust itself as needed. The modularity built into this design allows the development of flexible logic to control adaptability; through the definition of specific fidelity, strictness and parsimony methods, the system is able to dynamically adjust algorithmic parameters (such as $\tau_{dep}, \tau_{loop}, and, \tau_{sig}$), specifically as they relate to the topological characteristics of the current event stream. This is confirmed by the experimental outcomes that show the effectiveness of the change in architecture and that the concept of local optimization is no longer just an advantage but rather an imperative to address the mutually exclusive constraints that the standard global benchmark solutions were unable to address.
Furthermore, the testing of the Just-In-Time mechanism verified that it is indeed possible to dissociate the efficiency of the computation from the duration of the stream. Though the experiments on the payroll log showed an efficiency increase of 98%, the general implication is that the scalability of the optimization follows the law of the double-tap principle ($N_{opt} \approx 2 \times N_{drifts}$), which limits the use of the resource to the initiation and establishment of structural changes.

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